

Question :

Java programming -Java Buzzwords, Data types, Variables, Operators, Control structures, Arrays, Console input and output

Sample Program:

1. Sample Java program to check whether the given number is prime palindrome or not.
CO1.

1. Steps of the Program :

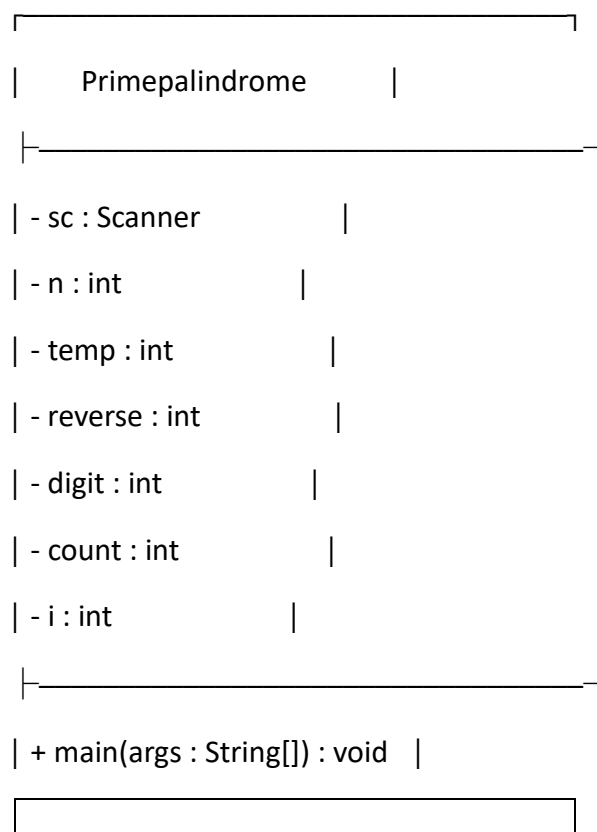
1. Import the Scanner class.
2. Create the Primepalindrome class.
3. Create a Scanner object to get input from the user.
4. Read the number n.
5. Store the value of n in temp.
6. Initialize reverse = 0.
7. Use a while loop to reverse the number.
8. Initialize count = 0.
9. Use a for loop to find the number of factors of n.
10. If count == 2, the number is prime.
11. If reverse == n, the number is a palindrome.
12. If both conditions are true, display **Prime Palindrome**.
13. Otherwise, display **Not a Prime Palindrome**.
14. Close the Scanner.
15. End the program.

2. Overall Algorithm

1. **Start**
2. Read an integer n.
3. Store n in temp.
4. Set reverse = 0.

5. Repeat while temp > 0:
 - Find the last digit using digit = temp % 10.
 - Add the digit to reverse.
 - Remove the last digit using temp = temp / 10.
6. Set count = 0.
7. Repeat i from 1 to n:
 - If n % i == 0, increment count.
8. Check whether reverse == n and count == 2.
9. If both are true, print "**n is a Prime Palindrome**".
10. Otherwise, print "**n is not a Prime Palindrome**".
11. **Stop.**

Uml Diagram :



Code :

```
import java.util.Scanner;
```

```
public class Primepalindrome {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter a number: ");  
        int n = sc.nextInt();  
        int temp = n;  
        int reverse = 0;  
  
        while (temp > 0) {  
            int digit = temp % 10;  
            reverse = reverse * 10 + digit;  
            temp = temp / 10;  
        }  
        int count = 0;  
  
        for (int i = 1; i <= n; i++) {  
            if (n % i == 0) {  
                count++;  
            }  
        }  
        if (reverse == n && count == 2)  
            System.out.println(n + " is a Prime Palindrome");  
        else  
            System.out.println(n + " is not a Prime Palindrome");  
  
        sc.close();  
    }  
}
```

}

Output :

```
PS C:\Users\yabhi\abhi.java> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages'
'-cp' 'C:\Users\yabhi\AppData\Roaming\Code\User\workspaceStorage\368729c680f0f32712f9803f080a0563\redhat.java\jdt_ws\abhi.java_92e0559b\bin'
'Primepalindrome'
Enter a number: 575
575 is not a Prime Palindrome
PS C:\Users\yabhi\abhi.java> ^C
PS C:\Users\yabhi\abhi.java>
PS C:\Users\yabhi\abhi.java> c; cd 'c:\Users\yabhi\abhi.java'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+S
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\yabhi\AppData\Roaming\Code\User\workspaceStorage\368729c680f0f32712f9803f080a0563\redhat.j
ava\jdt_ws\abhi.java_92e0559b\bin' 'Primepalindrome'
Enter a number: 101
101 is a Prime Palindrome
PS C:\Users\yabhi\abhi.java>
```